

REGULATORY READINESS INFRASTRUCTURE

Regulatory readiness for **Indian medtech.**

A workflow simplification layer for founders navigating CDSCO. This briefing is for the Venture Center regulatory team — context ahead of Wednesday's live demo.

DATE

12 May 2026

AUDIENCE

Venture Center regulatory team

FORMAT

Pre-read · 14 pages · A4 landscape



What you'll see on Wednesday

A founder uploads a pitch deck and works through a structured wizard. Output is an editable submission scaffold for a CDSCO application. Four touchpoints:

01 · INTAKE

Pitch deck upload

Pitch deck and product context.
Structured extraction surfaces ~22 fields.

● ● ● clearpath.in/intake

↑ Drop pitch deck

PDF · max 20MB

✓ cardiorhythm-deck.pdf · 4.2 MB



02 · READING

Readiness reading

Readiness tier, top gaps, recommended pathway.

● ● ● clearpath.in/assessment/...

READINESS TIER

Early pilot stage

Class C SaMD

Indicative · not a validated benchmark

Top gaps

- Clinical evidence — pilot only
- ISO 13485 not yet started
- ACP undefined (AI/ML)



03 · WIZARD

Tier B wizard

Six core + two conditional questions specific to MD-7 pathway. Prefilled where possible.

● ● ● clearpath.in/upgrade/.../wizard

B3 · PREDICATE DEVICES

Name up to 3 predicate devices similar to yours.

AliveCor KardiaMobile · CDSCO 2022

Add another...



04 · OUTPUT

Editable submission scaffold

12 sections of an MD-7 / MD-9 application scaffolded for expert review. Fully editable.

● ● ● clearpath.in/pack/...

SUBMISSION SCAFFOLD · MD-7

- ✓ §1 Cover letter
- ✓ §2 Constitution
- ✓ §3 Site documents
- §4 Plant Master File
- §5 QMS — pending
- ... 7 more sections

ClearPath structures readiness workflows. It does not replace regulatory consultants or guarantee approvals. Outputs are designed to support founder preparedness and expert review — not autonomous regulatory decision-making. Every section is editable and intended as a starting point for qualified regulatory experts.

Current scope of structured workflows

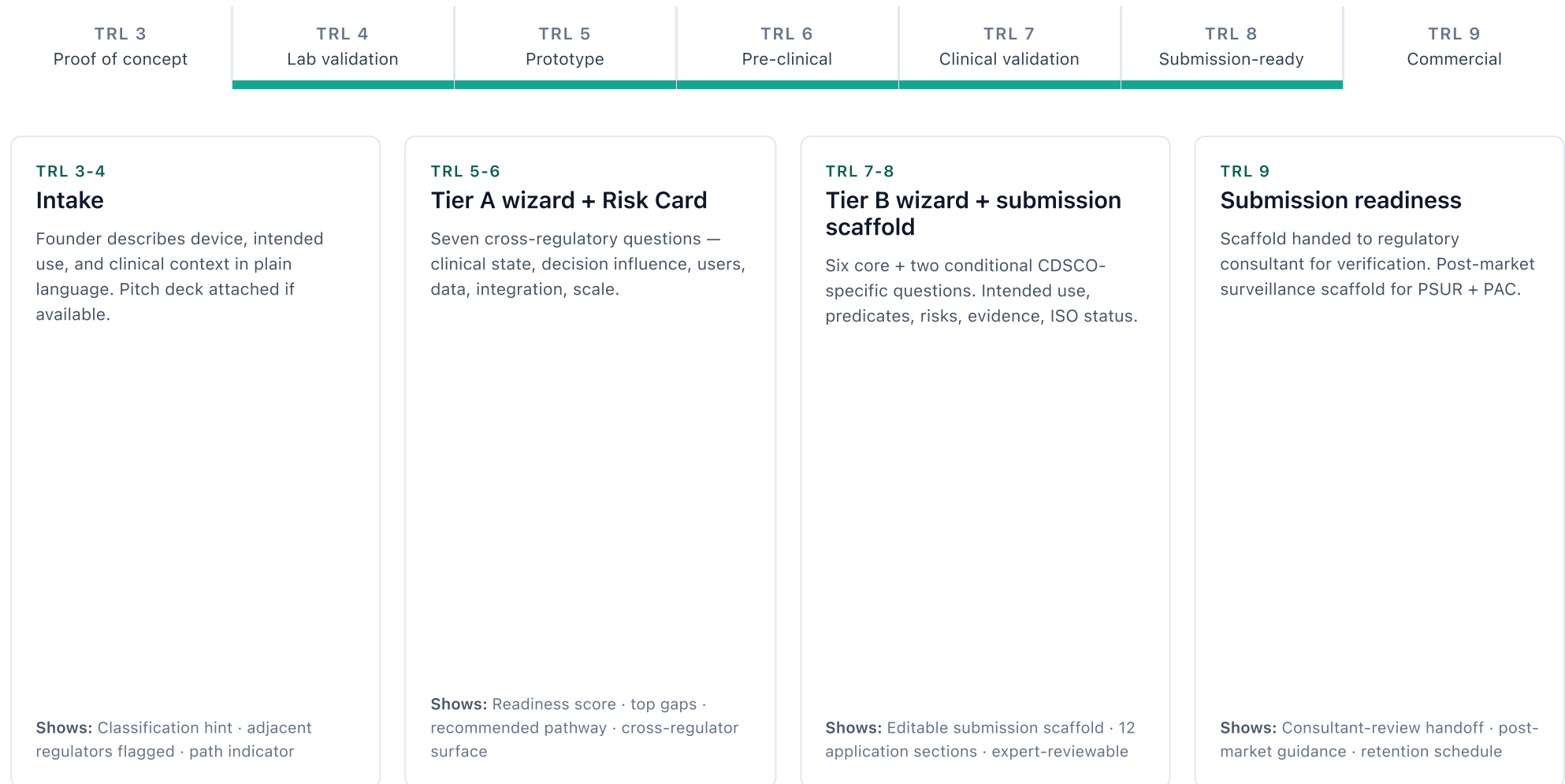
ClearPath does not cover all of Indian medtech regulation. The map below is honest about what's in product today versus what's scoped for later. Colour reflects status, not aspiration.

■ Production — supported end-to-end today
 ■ Experimental — partial logic, under validation
 ■ Sprint 3+ roadmap — known scope, not yet built

PRODUCTION Medical device — Class A / B / C / D mfg MD-3 → MD-5 (SLA) MD-7 → MD-9 (CLA) Full Tier A + Tier B submission scaffold for hardware MD manufacturing. Self-notification path documented.	PRODUCTION Software as Medical Device (SaMD) MD-3 / MD-7 + SaMD overlay Risk classification matrix + Oct 2025 Draft Guidance overlays. AI/ML lifecycle questions in Tier B.	EXPERIMENTAL In-vitro diagnostics (IVD) MD-3 / MD-7 + MD-24 / MD-25 MD-28 → MD-29 IVD pathway logic mapped in source document; explicit IVD-vs-MD routing question targeted for Sprint 3.	EXPERIMENTAL Clinical investigation (pre-commercial) MD-12 → MD-13 (test mfg) MD-22 → MD-23 (CI permission) MD-26 → MD-27 (no-predicate) Pre-trial test license pair (MD-12/13) and CI permission pair (MD-22/23) both modelled. Wizard surfacing in progress.
SPRINT 3+ ROADMAP Combination products PM0A-dependent DCG(I) joint review Documented; primary-mode-of-action question and DCG(I) handoff workflow pending.	SPRINT 3+ ROADMAP Import pathways MD-14 → MD-15 MD-16 → MD-17 (test) MD-18 / MD-20 (narrow) Authorised-agent + apostille + FSC workflow scoped; not yet productised in wizard.	SPRINT 3+ ROADMAP Pharma-linked workflows CT-04 / CT-04A / CT-06 NDCT 2019 Form 22 / 25 (D&C Act 1945) Outline-only in source document. Scope decision deferred to v2 — depends on cohort demand.	SPRINT 3+ ROADMAP Loan licence + personal import MD-4 → MD-6 / MD-8 → MD-10 MD-20 → MD-21 Specialist pathways. Documented for completeness; productisation prioritised by founder pipeline.

Workflow depth scales with TRL

A team at TRL 4 should not see the same wizard as a team at TRL 8. ClearPath uses progressive disclosure — the question set grows as evidence and decisions mature.



Representative founder journeys

Different device categories produce different document graphs. Four shapes illustrate the spread; the source document covers 14 distinct personas.

SaMD

AI/ML SaMD founder

Software-only product, clinical decision support

Class B or C (situation × significance)
Forms MD-7 → MD-9 + ACP overlay
Timeline ~6-9 months (no-predicate adds 12-24)

Pain: classification depends on intended use phrasing; adaptive AI/ML triggers Algorithm Change Protocol obligations that are still in draft.

MD

Implantable device startup

Class D hardware with biocompatibility + sterilisation

Class D (implant, high-risk)
Forms MD-7 → MD-9 (CLA)
Timeline 3-6 months published; longer in practice

Pain: extensive conditional documents (DMF §8.11 biocompatibility, §8.14 sterilisation, §8.18 clinical evidence) — easy to under-scope on first submission.

CI

Clinical investigation pathway

Two paired permissions, often confused

Pre-trial MD-12 → MD-13 test licence to manufacture
Trial MD-22 → MD-23 CI permission
Then MD-26 → MD-27 if no predicate

Pain: founders frequently apply for MD-22 without MD-13 in place to make trial devices. Sequencing is a common rejection cause.

IVD

IVD diagnostics startup

Separate appendix, in-country performance evaluation

Class A / B / C / D (own scale)
New IVD MD-24 → MD-25, MD-28 → MD-29
Then MD-3 / MD-7 / MD-14 marketing licence

Pain: Class B-D new IVDs need in-India CPE at registered MDTL even if foreign-approved. TB serology and malaria-antibody RDTs are prohibited categories — hard stops worth surfacing early.

Regulatory reasoning compression

~44 structured fields drive a graph of dozens of forms and documents. **Rules + ontology + workflow engine** — deterministic reasoning over CDSCO rule structure, not generative AI invention.

Founder input

Intake (7 fields) · Tier A (7 questions) · Tier B (6+2 questions) · structured pitch-deck extraction (~22 fields).

e.g. `intended_use`, `clinical_state`, `decision_influence`, `ai_ml`, `data_sensitivity`

Ontology mapping

Inputs mapped to MDR-2017 concepts: device class, predicate status, conditional triggers (sterile, patient-contact, drug, software, AI/ML, veterinary).

e.g. `decision_influence=drives` + `clinical_state=serious` → SaMD Class B

Workflow engine

Rules determine applicable form sequence, document checklist, conditional NOCs, and inter-form dependencies (e.g., MD-27 before MD-7).

e.g. `has_predicate=no` → MD-26 → MD-27 prerequisite for MD-7

Document graph

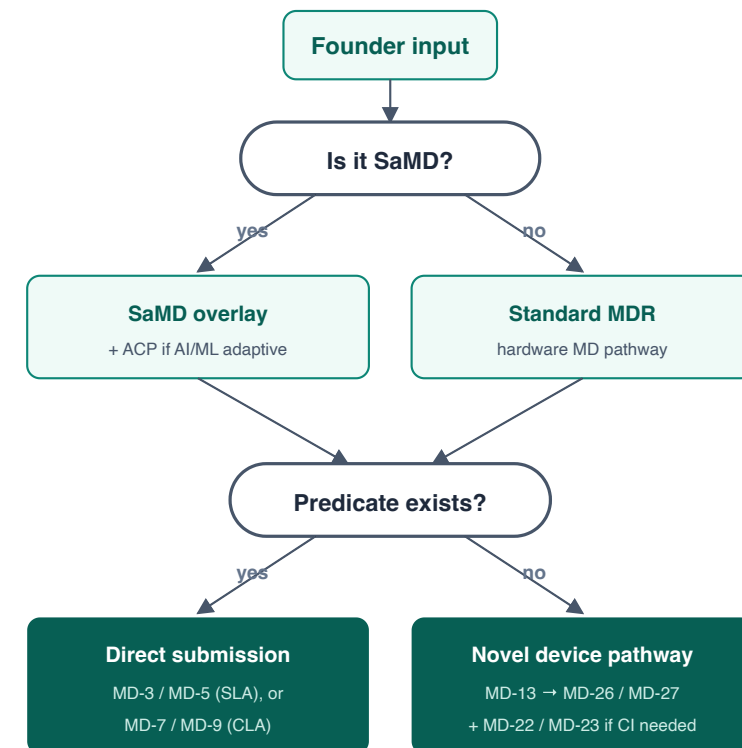
DMF, PMF, QMS, regulatory NOCs cross-referenced. ~60-80 distinct documents covered across persona set.

e.g. `sterile=yes` → DMF §8.14 sterilisation validation required

Workflow-assisted drafting

Per-section scaffold with explicit uncertainty flags. Founder-editable. Auditable

BRANCHING DECISIONS ENCODED IN RULES



Two of many decision branches encoded in the engine

A small slice of the deterministic decision tree. The same shape applies to dozens of other branches (sterile / patient-contact / drug content / veterinary / no-predicate).

Worked example — Class C AI/ML SaMD

CardioRhythm-style continuous cardiac event prediction. Six-step journey from upload to submission scaffold. Each step shows the actual surface the founder sees.

1 Pitch deck upload

clearpath.in/intake

Founder details

Dr Sneha R · sneha@...

Continuous ECG monitoring for early arrhythmia detection in post-op cardiac patients...

✓ cardiorhythm-deck.pdf · 4.2 MB

Founder fills 5 intake fields and uploads pitch deck. ~2 minutes.

2 Structured extraction

parsing...

Extracted fields · 22 of 22

device_class: samd_class_c_d
ai_ml: adaptive
data_sensitivity: identifiable
setting_of_use: inpatient
predicate: none identified

Pitch deck parsed via structured extraction. Suggested fields prefill the wizard; founder confirms or edits.

3 Tier A — prefilled

clearpath.in/wizard/.../q/1

◆ 4 of 7 prefilled from deck

TIER A · Q5 — INTEGRATION

Does your product integrate with ABDM, HIS, or hospital systems?

Wizard opens at first unanswered question. Founder reviews 4 prefills, answers 3 remaining. ~3 min.

4 Readiness reading

clearpath.in/assessment/...

READINESS TIER

Early pilot stage Class C SaMD

Readiness tier · indicative only

RECOMMENDED PATH

MD-13 → MD-26 → MD-27 → MD-7 → MD-9

no predicate identified · investigational MD path

First reading of where the founder stands. Surfaces the no-predicate sequence early.

5 Tier B — prefilled

clearpath.in/upgrade/.../wizard

◆ B1, B2, B5, B6 prefilled

TIER B · B3 — PREDICATE DEVICES

Name up to 3 predicate devices similar to yours.

AliveCor KardiaMobile · CDSO 2022

⚠ I have no predicate device

B1, B2, B5, B6 prefilled from pitch deck. Founder edits B1 wording, completes B3 + B4. ~10 min.

6 Submission scaffold

clearpath.in/pack/...

MD-7 SUBMISSION SCAFFOLD · 9 OF 12 SECTIONS READY

✓ §1 Intended use · §2 Class justification
✓ §3 Substantial equivalence
✓ §4 PMF · §5 QMS undertaking
✓ §6 Risk management · §7 V&V
● §8 ACP for AI/ML (draft)
○ §9 Clinical evidence — pivotal pending

12 sections scaffolded for expert review. ACP section flagged as draft pending finalised Oct 2025 guidance.

Why this matters for translational medtech ecosystems

The recurring pattern across BIG / TBI / incubator cohorts: technical depth is high; regulatory sequencing is low. ClearPath sits between proof-of-concept and consultant engagement.

- 1 **Founders discover regulation too late.** Device-class implications are often surfaced only after first investor diligence or first hospital pilot — by which point design and clinical-evidence choices are locked.
- 2 **TRL ladder and regulatory readiness are disconnected.** Technical TRL frameworks rarely surface MDR-2017 prerequisites at each stage; teams hit regulation as a wall rather than a parallel track.
- 3 **Teams may initiate MD-22 clinical investigation discussions before obtaining MD-13 test manufacturing permissions.** The sequencing dependency between MD-12/13 (test licence to manufacture trial devices) and MD-22/23 (CI permission) is a recurring source of rework.
- 4 **AI/ML SaMD founders often underestimate ACP / change-management documentation requirements.** The Oct 2025 draft guidance frames Algorithm Change Protocol as a five-component file. Teams shipping adaptive models discover this late.
- 5 **Consultant access doesn't scale to every cohort member.** Specialist regulatory consultants are gated by budget and time; programme-level support needs structured pre-consultant prep that is easier to triage.

INDIAN MEDTECH CONTEXT (SECONDARY)

~1,000 medical-device startups

broader medtech + health-tech: 3,000-6,000

BioSpectrum Asia; Accorp 2025

600+ devices under mandatory CDSCO registration

market ~\$12B FY24 → projected ~\$50B by 2050 (~15% CAGR)

ORF; Accorp 2025

BIG / incubator fit. BIRAC's BIG cohort scope explicitly includes medical devices, diagnostics, and drug-delivery — exactly the founders who hit MDR-2017 navigation at TRL 5-7. Venture Center is BIRAC's 3rd Regional Centre. venturecenter.co.in; birac.nic.in

Privacy, auditability, and governance posture

Founders share regulated information. Our posture is concrete where shipped and explicit where in-progress. DPDP Rules 2025 are now notified; substantive provisions phase in through May 2027.

 No model training on uploads Pitch decks and intake content are not used to train models. Anthropic's API non-training policy governs the model layer. <i>In place today</i>	 India-hosted data Application data on Supabase Mumbai region. Supports DPDP Act 2023 data-residency posture. <i>In place today</i>	 Time-bound retention Pitch decks and intake data purged 90 days after scaffold delivery. Per-customer override available. <i>Default policy; configurable</i>	 Audit logs Every scaffold generation logged with timestamp, input version, and engine version for traceability. <i>In place today</i>	 DPDP-aligned roadmap Board operational from 13 Nov 2025; substantive provisions effective by 13 May 2027. We are tracking the phased timeline. <i>In progress; on schedule</i>
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WHAT CLEARPATH IS NOT Not a regulatory approval platform · Not a substitute for notified bodies or regulatory consultants · Not legal advice

DPDP timeline: DLA Piper · Lexology · Rödl Insights (2025-2026 implementation rollout). Anthropic API non-training: anthropic.com/legal-commercial-terms.

UNCERTAIN flags — transparent by design

Where CDSCO rule text is silent, ambiguous, or in active draft, ClearPath surfaces uncertainty rather than fabricates confidence. The internal reference document carries 85 such flags; the most consequential are below.

IVD Master File appendix conflict

IVD FAQ §102 places contents in Appendix II of Fourth Schedule; multiple republished checklists reference Appendix III. Cascades through every IVD persona — needs authoritative ruling.

Algorithm Change Protocol — bridge regime

Oct 2025 SaMD Draft introduced ACP framework. Stakeholder comments closed Nov 2025; finalisation pending. What posture should an AI/ML SaMD shipping in 2026 take?

CTRI registration for medical device CI

Mandatory for pharma trials per NDCT 2019. MDR-2017 Seventh Schedule references protocol registration; whether CTRI specifically is required for medical device CI needs confirmation.

SEC composition and convening

Subject Expert Committee is referenced across novel-device, classification-dispute, and CI-waiver paths. Composition rules and scheduling cadence are not publicly codified.

Combination product PMOA determination

No published CDSCO procedure equivalent to FDA's Office of Combination Products determination. Borderline cases go to SEC — but the upstream decision rule is informal.

HOW UNCERTAINTY APPEARS IN PRODUCT



clearpath.in/pack/.../section/9

SUBMISSION SCAFFOLD · §8 — ALGORITHM CHANGE PROTOCOL

⚠ UNCERTAIN — ACP bridge regime

The Oct 2025 Draft Guidance introduces ACP for AI/ML SaMD. Draft is under comment; bridging requirements between now and finalisation are not yet codified. We recommend consulting CDSCO or a regulatory advisor before final submission.

The applicant proposes the following Algorithm Change Protocol covering data management, performance monitoring, retraining triggers, software update, and rollback...

Bible §6 Algorithm Change Protocol · §36.4 Q13

Uncertainty is shown in-line where it applies, with a reference back to the source bible section for the founder's consultant to follow up on.

Where your feedback would be invaluable

We are not asking for a stamp of approval. We are asking for the kind of correction and pattern-sharing that improves a regulator-facing tool meaningfully.

1 Edge cases we are missing.

e.g. a device-class transition pattern that recurs in your cohort

2 Workflow gaps in our persona coverage.

e.g. a founder shape from a recent BIG cohort that doesn't fit any of our 14 personas

3 Reviewer heuristics we should encode.

e.g. "if a Class C SaMD doesn't have a predicate, expect SEC referral on these grounds"

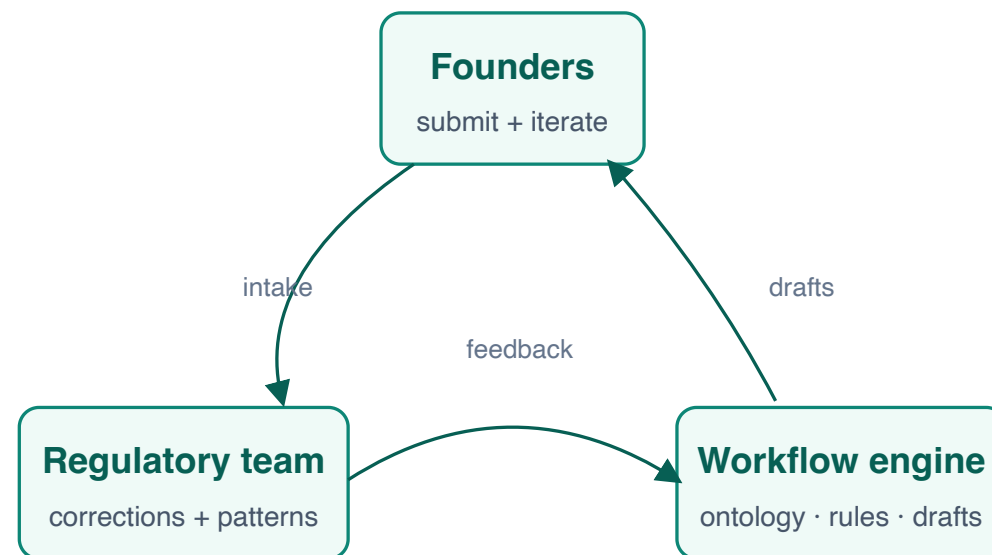
4 2-3 sample cohort pathways for end-to-end validation.

anonymised; we'd run ClearPath against them and share back the submission scaffold for your review

5 Interpretation corrections where our reading of MDR-2017 is wrong.

the source bible has 85 UNCERTAIN flags; the top 5 are on the previous slide

THE LOOP WE'RE TRYING TO CLOSE



A small amount of regulator-team feedback compounds across thousands of founder workflows. That is the asymmetry we are trying to use carefully.

Wednesday demo flow

Thirty minutes, structured to leave time for your questions. We can drill into any stage that warrants more attention.

#	STAGE	WHAT IT SHOWS	TIME
01	Pitch deck upload + automatic extraction	Founder uploads CardioRhythm deck. Watch structured extraction surface ~22 fields and pre-classify the device.	5 min
02	Tier A wizard end-to-end	Seven cross-regulatory questions with pre-fills from the deck. Manual editing to show audit trail.	10 min
03	Risk Card review	Readiness score, top gaps, recommended pathway. Cross-regulatory call-outs (DPDP, ABDM, AERB).	3 min
04	Tier B wizard — prefill + completion	Six core + two conditional questions. Predicate picker, risk-mitigation block, ISO 13485 status.	10 min
05	Submission scaffold output structure	Walk the 12-section submission scaffold. Highlight UNCERTAIN flags and how they appear in-line.	5 min
Q&A	Reserved for your team	Edge cases, sample cohort pathways, interpretation corrections — anything from slides 10-11.	— open

Founder + institutional context

Brief context on the team and the institutional positioning behind ClearPath.

Raunaq Pradhan

Founder, ClearPath

Building ClearPath as an India-focused regulatory readiness layer for medtech founders. Background spans Indian digital health, product strategy, and exposure to ABDM-era infrastructure design.

WHY THIS PROBLEM

Indian medtech founders consistently report the same friction: CDSCO sequencing is opaque until they are already invested in a path. Consultant access is gated. Cohort programmes cannot scale specialist time to every member. A structured pre-consultant readiness layer fills a real gap.

ADJACENT ECOSYSTEM FAMILIARITY

Working knowledge of ABDM consent framework, DPDP Act 2023 obligations, ICMR ethical guidelines for biomedical research, and the public CDSCO surface — MD Online portal, NSWS, BharatKosh.

BUILD POSTURE

Tier A wizard, Tier B wizard, and structured pitch-deck extraction in production today. Persona branching (IVD, CI, combination) and operational details (EC, sample size, fees, timelines) are scoped on top of the comprehensive reference document for upcoming iterations.

INSTITUTIONAL POSTURE

Built as independent regulatory infrastructure for the Indian medtech ecosystem. Open to incubator partnerships, cohort pilots, and institutional collaboration with bodies actively supporting translational medtech.

EMAIL

raunaq.pradhan@gmail.com

Extended reference architecture

The briefing above is a compressed read of a much larger internal reference. Available on request.

WHAT'S IN THE REFERENCE

14

PERSONAS

24+

FORMS COVERED

7

DEVICE TAXONOMIES

85

UNCERTAIN FLAGS

Three formats. Markdown source (3,800 lines) · single-file HTML (sidebar TOC, shareable anchors) · 102-page A4 PDF for offline reading. All derived from one canonical source.

CLOSING

Designed to reduce avoidable regulatory friction for Indian medtech founders.

ClearPath · regulatory readiness infrastructure
Briefing prepared for Venture Center regulatory team · 12 May 2026
Full reference document available on request.